



LINEAR ALGEBRA AND ITS APPLICATIONS

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CLASS-8

POWERS AND PRODUCTS OF MATRICES

Computation of Powers of a square matrix:

Diagonalization of a square matrix A also helps to find the Powers of $A, A^2, \dots$ etc.

$$\text{We have } D = S^{-1}AS$$

$$D^2 = (S^{-1}AS)(S^{-1}AS)$$

$$D^2 = S^{-1}A^2S$$

Pre-multiplying by S and Post-multiplying by S^{-1}

$$SD^2S^{-1} = S\tilde{S}^{-1}A^2S\tilde{S}^{-1} = I A^2 I = A^2$$

$$A^2 = S D^2 S^{-1}$$

In general, $A^n = S D^n S^{-1}$

where $D^n = \begin{bmatrix} \lambda_1^n & 0 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2^n & 0 & 0 & \dots & 0 \\ 0 & 0 & \lambda_3^n & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & \lambda_n^n \end{bmatrix}$

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Powers and Products of matrices



Problems:-

1. Diagonalize the matrix $A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$ and find

one of its square roots. How many square roots

will be there?

Ans: $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 5-\lambda & 4 \\ 4 & 5-\lambda \end{vmatrix} = 0 \Rightarrow (5-\lambda)^2 - 16 = 0$
 $\Rightarrow \lambda = 1, 9$

consider $[A - \lambda I]x = 0$

$$\Rightarrow (5-\lambda)x + 4y = 0$$

$$4x + (5-\lambda)y = 0$$

Case 1:- When $\lambda = 1$

$$4x + 4y = 0 \quad \therefore X_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$
$$4x + 4y = 0$$

Case 2:- When $\lambda = 9$

$$-4x + 4y = 0 \quad \therefore X_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$
$$4x - 4y = 0$$

$$\therefore S = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 9 \end{bmatrix}, \quad S^{-1} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

We know that $A = S \Lambda S^{-1}$

$$A^{Y_2} = S \Lambda^{Y_2} S^{-1}$$

$$= \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 9 \end{bmatrix}^{Y_2} \begin{bmatrix} -\frac{Y_2}{2} & \frac{Y_2}{2} \\ \frac{Y_2}{2} & \frac{Y_2}{2} \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -\frac{Y_2}{2} & \frac{Y_2}{2} \\ \frac{Y_2}{2} & \frac{Y_2}{2} \end{bmatrix}$$

$$A^{Y_2} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

We have 4 square roots $\therefore \Lambda^{Y_2} = \sqrt{1} = \pm 1$ & also $\sqrt{9} = \pm 3$.

We get different values when we consider $\pm 1, \pm 3$.

LINEAR ALGEBRA AND ITS APPLICATIONS

Powers and Products of matrices

② Diagonalize $A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$ & hence find A^{100} .

Show that $A^{100} \equiv A$.

Ans: $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} \frac{1}{2} - \lambda & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} - \lambda \end{vmatrix} \Rightarrow \begin{matrix} \lambda^2 - \lambda = 0 \\ \lambda = 0, \lambda = 1 \end{matrix}$

consider $[A - \lambda I]x = 0 \Rightarrow \begin{pmatrix} \frac{1}{2} - \lambda \end{pmatrix}x + \frac{1}{2}y = 0$
 $\frac{1}{2}x + \left[\frac{1}{2} - \lambda\right]y = 0$

Case 1:- When $\lambda = 0 \Rightarrow \frac{x}{2} + \frac{y}{2} = 0 \Rightarrow x = -y \quad x_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$
 $\frac{x}{2} + \frac{y}{2} = 0$

Case 2: When $\lambda = 1$

$$-\frac{x}{2} + \frac{y}{2} = 0 \Rightarrow x = y \quad x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\frac{x}{2} - \frac{y}{2} = 0$$

$$A = S \Lambda S^{-1}$$

$$A = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$A^{100} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}^{100} \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$A^{100} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = A$$



THANK YOU

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